

Branch: B.Tech (Chemical/Petrochemical)

Subject with Subject Code: Mass Transfer Operation-I (BTCHC502)

Date: -11/12/19

Semester: V

Marks: 60

Time: 3 Hrs.

Instructions to the Students

- Each question carries 12 marks.
- Attempt **any five** questions of the following.
- Illustrate your answers with neat sketches, diagram etc., wherever necessary.
- If some part or parameter is noticed to be missing, you may appropriately assume it and mention it clearly.

(Marks)

Q.1.	a	In an air-carbon dioxide mixture at 298 K and 202.6 kPa, The concentration of CO ₂ at two planes (3 mm) apart are 15 vol.% and 25 vol.%. The diffusivity of CO ₂ in air at 298 K and 202.6 kPa is $8.2 \times 10^{-6} \text{ m}^2/\text{s}$. Calculate the rate of transfer of CO ₂ across the two planes, assuming: i. Equimolecular counter diffusion. ii. Diffusion of CO ₂ through a stagnant air layer.	4																		
	b	Give the Differences between two fluxes J_A and N_A . Write down the relationship between mass transfer coefficients and diffusivity with details.	4																		
	c	Describe in short the phenomena of equimolar counter diffusion and diffusion of one component through non diffusing other component with applications from Chemical Engg as well as real life.	4																		
Q.2.	a	The equilibrium adsorption of benzene vapor on certain activated charcoal at 33 °C is reported as follows. <table border="1"><thead><tr><th>Benzene vapor adsorbed cm³/gm charcoal</th><th>Partial pressure benzene in mm Hg</th></tr></thead><tbody><tr><td>15</td><td>0.001</td></tr><tr><td>25</td><td>0.0045</td></tr><tr><td>40</td><td>0.0251</td></tr><tr><td>50</td><td>0.115</td></tr><tr><td>65</td><td>0.251</td></tr><tr><td>80</td><td>1.00</td></tr><tr><td>90</td><td>2.81</td></tr><tr><td>100</td><td>7.82</td></tr></tbody></table> Charcoal which has adsorbed upon it 100 cm ³ of benzene vapor per gm is to be stripped	Benzene vapor adsorbed cm ³ /gm charcoal	Partial pressure benzene in mm Hg	15	0.001	25	0.0045	40	0.0251	50	0.115	65	0.251	80	1.00	90	2.81	100	7.82	8
	Benzene vapor adsorbed cm ³ /gm charcoal	Partial pressure benzene in mm Hg																			
15	0.001																				
25	0.0045																				
40	0.0251																				
50	0.115																				
65	0.251																				
80	1.00																				
90	2.81																				
100	7.82																				

2

	at the rate of 45.5 kg/hr of its benzene content to a concentration of 5% adsorbed benzene per gm of charcoal by continuous countercurrent contact with a stream of pure nitrogen gas at 1 std atm. The temperature will be maintained at 33 °C.																																		
	(i) What is minimum rate of nitrogen flow in kg/hr?																																		
	(ii) What are number of ideal stages if nitrogen is 2 time minimum?																																		
b	For an interphase mass transfer between gas and liquid phase, derive relation between overall and individual k type mass transfer coefficients w.r.t. gas phase and liquid phase. Use usual terminology.	4																																	
3. a	A coal gas is to be freed of its light oil by scrubbing with wash oil as an absorbent and the light oil recovered by stripping the resulting solution with steam. The circumstances are as follows: Absorber: Gas in, 0.250 m³/s at 26°C, P _t = 1.07*10 ⁵ N/m², containing 2.0% by volume of light oil vapors. The light oil will be assumed to be entirely benzene, and a 95% removal is required. The wash oil is to enter at 26 C, containing 0.005 mole fraction benzene and has an average molecular weight 260. An oil circulation rate of 1.5 times the minimum is to be used. Wash oil-benzene solutions are ideal. The temperature will be constant at 26°C. At 26 °C, the vapor pressure of benzene = 13330 N/m². Compute the oil circulation rate.	9																																	
b	Explain selection criteria for choice of solvent for absorption.	3																																	
4. a	Oil is extracted from crushed seeds by a low boiling hydrocarbon. The following experimental data are collected on the concentration and solid content of the underflow and solid content of the overflow and the solution retained in the underflow. The concentrations of both the streams are equal on solid free basis.	8																																	
<table><tr><th colspan="2">Overflow</th><th>Underflow</th></tr><tr><th>kg oil per kg clear solution</th><th>kg solid per kg solution</th><th>kg inerts per kg solution</th></tr><tr><td>0.0</td><td>0.0</td><td>0.67</td></tr><tr><td>0.05</td><td>0.002</td><td>0.66</td></tr><tr><td>0.2</td><td>0.005</td><td>0.64</td></tr><tr><td>0.25</td><td>0.007</td><td>0.625</td></tr><tr><td>0.30</td><td>0.01</td><td>0.60</td></tr><tr><td>0.35</td><td>0.013</td><td>0.58</td></tr><tr><td>0.40</td><td>0.017</td><td>0.55</td></tr><tr><td>0.45</td><td>0.022</td><td>0.51</td></tr><tr><td>0.50</td><td>0.029</td><td>0.46</td></tr></table>			Overflow		Underflow	kg oil per kg clear solution	kg solid per kg solution	kg inerts per kg solution	0.0	0.0	0.67	0.05	0.002	0.66	0.2	0.005	0.64	0.25	0.007	0.625	0.30	0.01	0.60	0.35	0.013	0.58	0.40	0.017	0.55	0.45	0.022	0.51	0.50	0.029	0.46
Overflow		Underflow																																	
kg oil per kg clear solution	kg solid per kg solution	kg inerts per kg solution																																	
0.0	0.0	0.67																																	
0.05	0.002	0.66																																	
0.2	0.005	0.64																																	
0.25	0.007	0.625																																	
0.30	0.01	0.60																																	
0.35	0.013	0.58																																	
0.40	0.017	0.55																																	
0.45	0.022	0.51																																	
0.50	0.029	0.46																																	

3

		(i) Calculate and plot the equilibrium data on a suitable diagram. (ii) For the above problem, if 1000 kg of crushed seeds containing 25 % oil and 75 % inerts in intimately mixed with 1800 kg of the hydrocarbon and adequate time for settling is allowed. Calculate mass of underflow, mass of overflow and fraction of the oil removed in the extract.																																	
b		Explain the procedure in brief how to find number of ideal stages in continuous countercurrent extraction (of type I) when solvent rate is given for given feed rate. Use triangular or rectangular or solvent free system for explanation.	4																																
Q.5. a		Equilibrium adsorption data for propane on activated carbon at two different temperatures are given below. Fit the data using Langmuir isotherm & answer the following questions. What is the correlation coefficient (q_m) at temperature 310.9 K ? What is the correlation coefficient (K) at temperature 310.9 K ? What is the correlation coefficient (q_m) at temperature 366.5 K ? What is the correlation coefficient (K) at temperature 366.5 K	6																																
<table border="1"> <thead> <tr> <th colspan="2">$T=310.9 \text{ K}$</th><th colspan="2">$T=366.5 \text{ K}$</th></tr> <tr> <th>$p \text{ (kPa)}$</th><th>$q \text{ (m mol/g)}$</th><th>$p \text{ (kPa)}$</th><th>$q \text{ (m mol/g)}$</th></tr> </thead> <tbody> <tr><td>2.27</td><td>1.044</td><td>21.07</td><td>1.677</td></tr> <tr><td>15.6</td><td>2.819</td><td>45.33</td><td>2.386</td></tr> <tr><td>31.74</td><td>3.48</td><td>93.34</td><td>2.954</td></tr> <tr><td>89.74</td><td>4.207</td><td>279.2</td><td>3.584</td></tr> <tr><td>293</td><td>4.9</td><td>461.9</td><td>3.922</td></tr> <tr><td>479.2</td><td>5.294</td><td>634.3</td><td>4.244</td></tr> </tbody> </table>			$T=310.9 \text{ K}$		$T=366.5 \text{ K}$		$p \text{ (kPa)}$	$q \text{ (m mol/g)}$	$p \text{ (kPa)}$	$q \text{ (m mol/g)}$	2.27	1.044	21.07	1.677	15.6	2.819	45.33	2.386	31.74	3.48	93.34	2.954	89.74	4.207	279.2	3.584	293	4.9	461.9	3.922	479.2	5.294	634.3	4.244	
$T=310.9 \text{ K}$		$T=366.5 \text{ K}$																																	
$p \text{ (kPa)}$	$q \text{ (m mol/g)}$	$p \text{ (kPa)}$	$q \text{ (m mol/g)}$																																
2.27	1.044	21.07	1.677																																
15.6	2.819	45.33	2.386																																
31.74	3.48	93.34	2.954																																
89.74	4.207	279.2	3.584																																
293	4.9	461.9	3.922																																
479.2	5.294	634.3	4.244																																
b		Write a short note on break through curve and mass transfer zone MTZ in adsorption.	3																																
c		Explain application of Freundlich equation for adsorption with example on one stage and two stage adsorption.	3																																
Q.6. a		Explain the following terms in brief with respect to tray towers (0.5X6) (i) Flooding (ii) Weeping (iii) Coning (iv) Overall efficiency (v) Point efficiency (vi) Murphree tray efficiency	3																																
b		Describe sparged vessels and its applications in brief.	3																																
c		Describe packed towers and its applications in brief.	3																																
d		Describe construction and working of spray tower.	3																																